InterCellar empowers lab-scientists in the downstream analysis of cell-cell communication from single-cell transcriptomics

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Background

Multiple algorithms are available to predict cell-cell communication (CCC) in single-cell RNA sequencing (scRNA-seq) data

Downstream analysis and biological interpretation of the output of these algorithms (CCC data) are hindered by the complexity of the results and often require programming skills alongside biological expertise

We developed InterCellar as an interactive platform to explore and analyze CCC data, with no programming skills required

InterCellar's workflow offers customization & filtering options, multiple visualizations, automatic functional annotation and download of results

Both customized and data-driven analyses guide scientists in the interpretation of CCC and in the generation of hypotheses

InterCellar is an R/Shiny application distributed as a Bioconductor package (https://bioconductor.org/packages/InterCellar/)

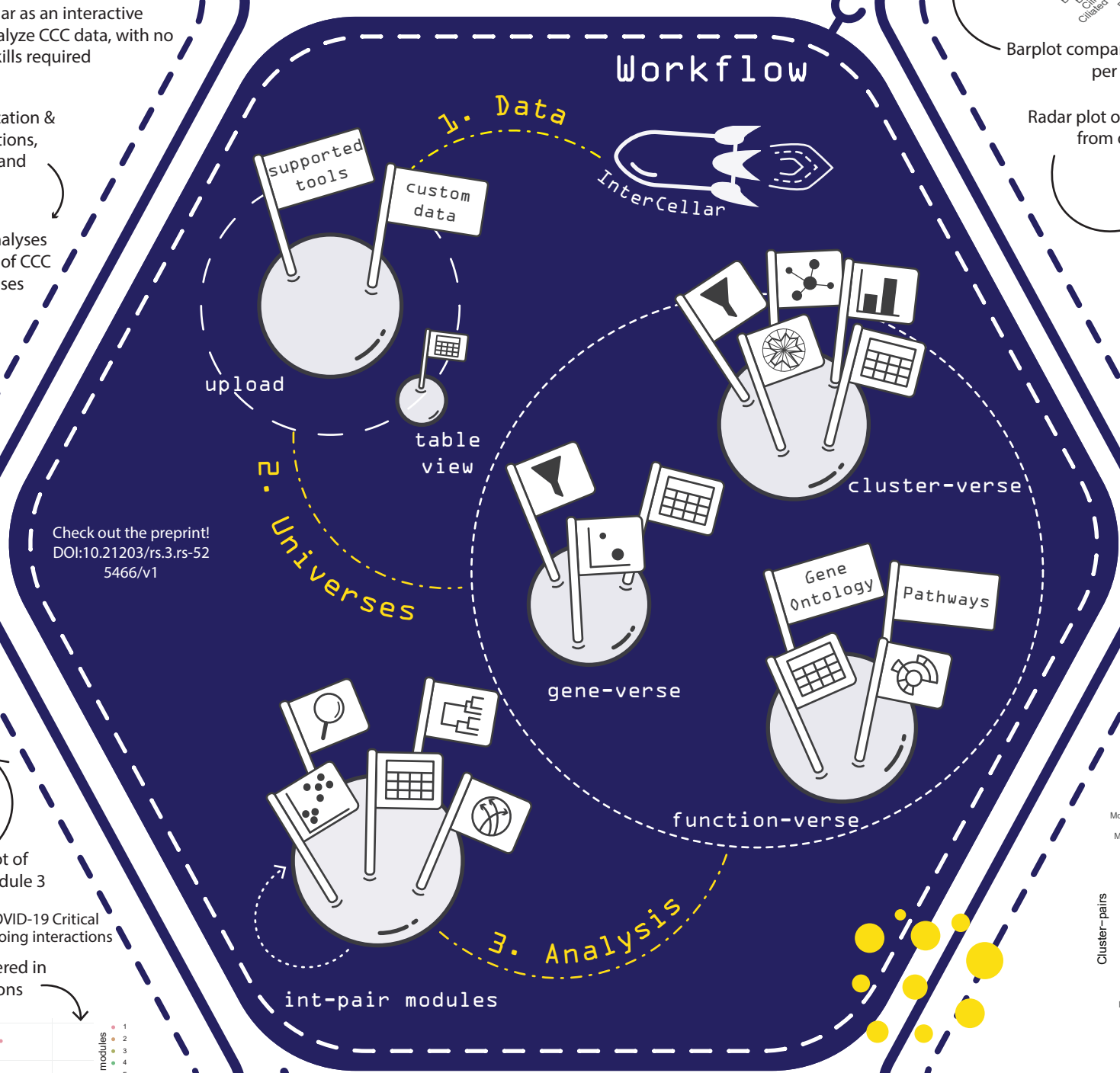
Upload data

Automatically, from supported tools:
- CellPhoneDB [1]
- SingleCellSignalR [2]
- ... more coming soon!

With your own custom CCC data, which contains:
- int-pairs (e.g. ligand-receptor genes)
- clusters / cell types
- molecular type of int-pairs (L or R)
- value (e.g. interaction score)

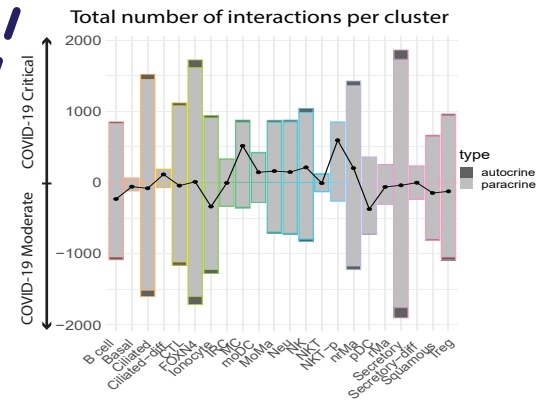
Here we use CellPhoneDB-generated CCC data from a published scRNA-seq dataset of COVID-19 patients divided in moderate and critical cases [3]

Workflow



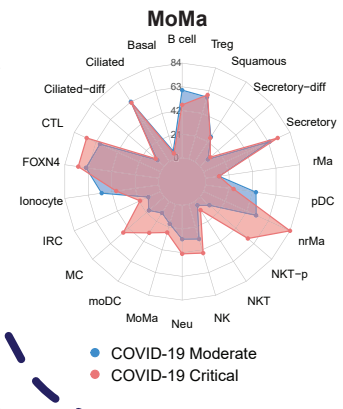
Check out the preprint!
DOI:10.21203/rs.3.rs-525466/v1

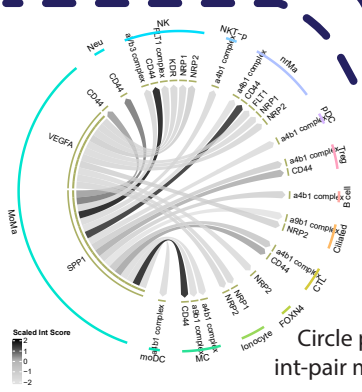
Cluster-verse



Barplot comparing the total number of interactions per cluster, in two conditions

Radar plot of the relative number of interactions from one cluster towards all others, in two conditions

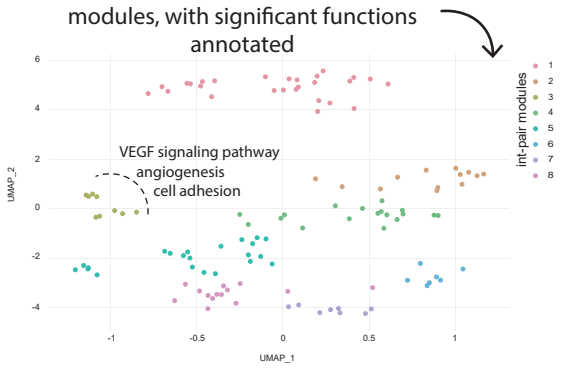




Circle plot of int-pair module 3

COVID-19 Critical
MoMa outgoing interactions

Interactive UMAP of int-pairs, clustered in modules, with significant functions annotated

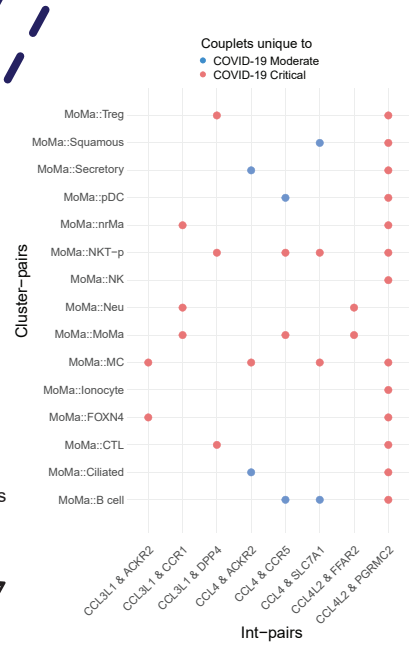


Focused analysis from a viewpoint cluster and a communication flow (directed-outgoing / directed-incoming /undirected)

Each defined as a subset of int-pairs with similar biological functions

Int-pair modules

Gene-verse

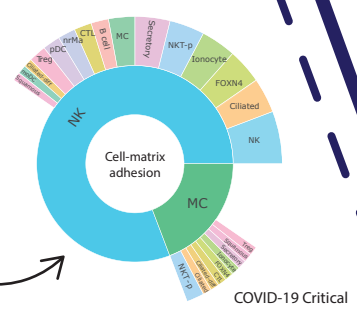


Selected CCL-pairs MoMa ->

Dotplot of selected int-pairs from the CC-chemokine subfamily showing occurrences of interactions between MoMa and other cell types. Only unique int-pair/cluster-pair couplets are shown, in the comparison of two conditions

Function-verse

Functional annotation of int-pairs:
- Gene Ontology
- KEGG
- Reactome
- Biocarta
- PID:NCI-Nature
- Panther
- PharmGKB



Interactive sunburst plot shows cluster-pairs communicating through int-pairs annotated to the GO functional term "cell-matrix adhesion"

InterCellar Features Legend

table, network, dotplot, viewpoint + flow, UMAP, filter, barplot, sunburst plot, dendrogram, circle plot

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References

[1] Efremova, M., Vento-Tormo, M., Teichmann, S. A. & Vento-Tormo, R. CellPhoneDB: inferring cell–cell communication from combined expression of multi-subunit ligand–receptor complexes. Nat. Protoc. 15, 1484–1506 (2020).
[2] Cabello-Aguilar, S. et al. SingleCellSignalR: inference of intercellular networks from single-cell transcriptomics. Nucleic Acids Res. 48, e55 (2020).
[3] Chua, R. L. et al. COVID-19 severity correlates with airway epithelium-immune cell interactions identified by single-cell analysis. Nat. Biotechnol. 38, 970–979 (2020).